

MID Semester of 3rd Sem B.Tech.

MID –SEMESTER EXAMINATION, SEPTEMBER, 2024

COURSE CODE: EAMTC04 (Backlog)

COURSE TITLE: MATHEMATICS FOR MACHINE LEARNING

TIME: 1.30 HRS

MAX. MARK: 25

Note:- Attempt all questions. Missing data/ Information (if any), may be suitably assumed & mentioned in the answer.

Q. NO	QUESTION	MARKS	CO
1a	$T: R^3 \rightarrow R^2$ defined by $T(x, y, z) = (y + z, y - z)$. Determine the matrix of the linear transformation T with respect to the standard ordered basis.	3	CO1
1b	Examine if the set $W = \{(x, y, z, w) \in R^4 : w = 2\}$ is a subspace of R^4 .	2	CO1
2a	Check whether $S = \{(1, 0, 1), (1, 1, 0), (1, 2, 3)\}$ of R^3 is linearly independent or linearly dependent.	3	CO1
2b	Find KerT of the linear mapping $T: R^3 \rightarrow R^2$ defined by $T(x, y, z) = (x - y, 2z)$.	2	CO1
3a	Examine whether T is a linear mapping where $T: R^2 \rightarrow R^2$ defined by $T(x, y) = (2x + y, x)$.	3	CO1
3b	If $\langle x + y, x - y \rangle = 0$, prove that $\ x\ = \ y\ $ vectors x, y are in real vector space with real inner product.	2	CO2
4a	Check whether the quadratic form $x^2 + 2y^2 - 4xz - 4yz + 7z^2$ is positive definite or positive semi-definite.	3	CO2
4b	Find Eigen values of the matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$	2	CO2
5a	Use Gram-Schmidt process to convert the basis $\{(1, 2 - 2), (2, 0, 1), (1, 1, 0)\}$ of the Euclidean space R^3 with standard inner product into an orthogonal basis.	4	CO2
5b	Examine whether the set $\left\{\left(0, \frac{1}{2}, 0\right), \frac{3}{2}(-1, 0, 1)\right\}$ is orthonormal set.	1	CO2